



UNITED INTERNATIONAL UNIVERSITY

COURSE SYLLABUS

1	School	School of Science & Engineering
2	Department	Department of CSE
3	Program	BSCSE [BSc in Computer Science & Engineering]
4	Name of Course	Probability and Statistics
5	Course Code	MATH 2205
6	Trimester and Year	Fall, 2025
7	Pre-requisites	Fundamental Calculus (MATH 1151)
8	Status	Supporting CSE Courses
9	Credit Hours	3.00
10	Section	E, G, O, and P
11	Class Hours	Saturday and Tuesday: 01:50 PM - 03:10 PM (Section E) Saturday and Tuesday: 11:10 AM - 12:30 PM (Section G) Sunday and Wednesday: 03:10 PM - 04:30 PM (Section O) Sunday and Wednesday: 11:10 AM - 12:30 PM (Section P)
12	Class Location	Room: 0702 (Section E) Room: 0707 (Section G) Room: 0632 (Section O) Room: 0702 (Section P)
13	Course website	http://elms.uiu.ac.bd/
14	Name (s) of Academic staff / Instructor(s)	Mahtab Uddin (MUn) Associate Professor – Mathematics Institute of Natural Sciences, UIU
15	Contact	mahtab@ins.uiu.ac.bd , 01550605560
16	Office	Room # 635/A
17	Counselling Hours	See eLMS.
18	Text Book	Probability and Statistical Inference by Robert V. Hogg, Elliot A. Tanis, and Dale L. Zimmerman, Pearson Education Asia, 9th Edition, 2015. Probability and Statistics for Engineers & Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, and Keying Ye Pearson Education Asia, 9th Edition, 2012.
19	Reference	Mathematical Statistics and Data Analysis by John A. Rice, International Thomson Publishing, 3rd Edition, 2007.
20	Equipment & Aids	Bring your own device (Any standard smartphone or tablet or laptop) to participate effectively in classroom activities. You are not allowed to borrow from others inside the classroom during class activities.
21	Course Rationale	This course helps to design data collection plans, analyze data appropriately and interpret and draw conclusions from those analyses. It also equips students with consequently requisite quantitative skills that they can employ and build on in flexible ways.
22	Course Description	Frequency distribution. Mean, median, mode and other measures of central tendency. Standard deviation and other measures of dispersion. Moments, skewness and kurtosis, correlation and regression analysis. Elementary probability theory. Mathematical Expectation, discontinuous probability distribution, e.g., binomial, Poisson and negative binomial. Continuous probability

		distributions, e.g., normal, exponential, and Gamma. Interval estimation, Hypothesis testing.			
23	Course Objectives	The course is designed to provide the background of the following topics 1. Define statistical data and focus on how to use them. 2. Analyze experimental and collected data using statistical method. 3. Explain description, interpretation and exploratory analysis of data by graphical and other means. 4. Discuss the fundamental probability theory and understand statistical reasoning and inferential methods. 5. Illustrate stochastic behavior of natural phenomena.			
24	Learning Outcomes	After the end of this course, the students will be able to: 1. Analyze data using measures of central tendency, measures of dispersion, point estimation, interval estimation and hypothesis testing. 2. Compute probability of dependent and independent events. 3. Develop probabilistic models for discrete and continuous random variables. 4. Demonstrate relation between multiple random variables. 5. Assess confidence intervals, decisions on hypothesis testing.			
25	Teaching Methods	Lecture, Question-Answer, Multimedia Projector.			
26	Topic Outline				
	Class	Topics Or Assignments	CLOs	Reading Reference	Activities
	1-4	Frequency distribution, measures of central tendency, measures of dispersion. Moments, skewness and kurtosis.	1	Chapter 1 Probability and Statistics for Engineers & Scientists	Question-Answer
	5-6	Correlation and regression analysis.	4	Chapter 11 Probability and Statistics for Engineers & Scientists	Question-Answer, Class Test-1
	7-10	Elementary probability theory, Conditional probability, Bayes' theorem.	2	Chapter 1 Probability and Statistical Inference	Question-Answer, Assignment-1, Class Test-2
	11-15	Discrete Random Variables and discrete distributions.	3	Chapter 2 Probability and Statistical Inference	Question-Answer
	16-19	Continuous Random Variables and continuous distributions.	3	Chapter 3 Probability and Statistical Inference	Question-Answer, Class Test-3
	20-21	Interval estimation.	5	Chapter 7 Probability and Statistical Inference	Question-Answer, Assignment-2, Class Test-4
	22-24	Hypothesis testing, Analysis of variances.	5	Chapter 8 Probability and Statistical Inference	Question-Answer, Class Test-5

27	Assessment Methods	Assessment Type					Mark
		Attendance					5%
		Class Tests					20%
		Assignments					5%
		Mid Term Exam					30%
		Final Exam					40%
28	Grading Policy	Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
		A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
		A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
		B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
		B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
		B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
					F (Fail)	<55	0.00
29	Additional Course Policies	1. Class Attendance and Participation: Class attendance is mandatory (at 80% of classes) to qualify for grading as per university policy. But I will grade you on the basis of your in-time presence. So after taking attendance of the class (usually at the beginning of the class), there will be no provision for recording attendance. Your in-time presence will also be considered as positive class participation. 2. Examination: There is NO provision for the make-up of missed classes and quizzes. Expect a quiz on the completion of each topic. 3. Assignment Failure to submit the Assignments on the due date will result in a 50% deduction from the possible score. 4. Counseling: You are expected to follow the counseling time-table as set out in this course.					
30	Additional Info	1. Academic Calendar: http://www.uiu.ac.bd/academic/calendar/ 2. Academic Information and Policies: http://www.uiu.ac.bd/academic/academic-information-policies/ 3. Grading and Performance Evaluation: http://www.uiu.ac.bd/academic/grading-performance-evaluation/ 4. Proctorial Rules: http://www.uiu.ac.bd/academic/1192-2/					